A force pair belongs to two different objects

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Interaction pair / ES: par de interacción / PT: par de interação

When A pushes B, B pushes A at the same time.

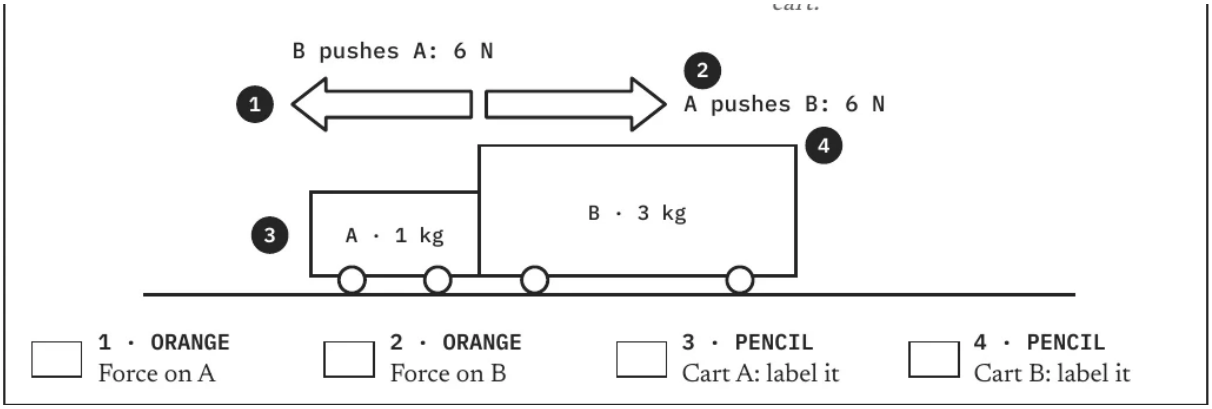
Also on your note page: Equal and opposite, Force receiver, Mass and acceleration.

B READ THE RELATIONSHIP

An interaction pair has two equal and opposite forces on different objects.

Swap the source and receiver to find the partner

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

Earth pulls an apple downward. Name the third-law partner force and the object that receives it.

Given Earth → apple gravity, down

Rule swap the object names

Work apple → Earth gravity, up

Answer Apple pulls Earth up Earth receives it

E SAVE FOR THE FINAL MEETING

When Earth pulls on XJ, XJ pulls on Earth just as hard. The forces are equal, but XJ's far smaller mass means its motion is the one that changes noticeably.

Read the physics: Interaction pair

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

If A pushes B, B pushes A at the same instant. The forces have the same strength. Swap the names to find the other receiver.

02 HOW IT WORKS

Draw one diagram for each receiver. Put only the force on that receiver in its diagram. These two arrows cannot cancel on one object.

03 WHAT TO CHECK

For the same force size, the smaller mass has the larger acceleration. Balanced forces on one object are a different idea.

04 TRY THE EXAMPLE FROM YOUR NOTES

Earth pulls an apple downward. Name the third-law partner force and the object that receives it. Apple pulls Earth up Earth receives it.

05 CONNECT TO XJ

When Earth pulls on XJ, XJ pulls on Earth just as hard. The forces are equal, but XJ's far smaller mass means its motion is the one that changes noticeably.

SAY IT BEFORE YOU WRITE

Do your two arrows end up on the same object? ___ pushes on ___. Its partner is ___ pushing on ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Interaction pair

ES: par de interacción / PT: par de interação

Meaning in my words	How can you tell?
How would you explain interaction pair to a friend? In my words, this means ___.	Earth pulls an apple down. What is the partner force, and which object feels it? I can tell because ___.
Example or sketch	**Non-example and why**
How would you label the pair when a hand pushes a cart? My example shows ___.	Are the table's push and Earth's pull on one book a third-law pair? I would change ___ because ___.

Equal and opposite

ES: iguales y opuestas / PT: iguais e opostas

Meaning in my words	How can you tell?
What does "equal and opposite" tell you about a pair of interaction forces? In my words, this means ___.	One skater pushes another with 30 N east. What force does the first skater feel from the second? I can tell because ___.
Example or sketch	**Non-example and why**
Draw the arrows for two carts whose bumpers press together. My example shows ___.	Do equal arrows guarantee equal accelerations for a heavy and a light skater? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Force receiver

ES: objeto que recibe la fuerza / PT: objeto que recebe a força

Meaning in my words	How can you tell?
How would you explain force receiver to a friend? In my words, this means ___.	In “the wall pushes on my hand,” where does that force arrow belong? I can tell because ___.
Example or sketch	**Non-example and why**
When my hand pushes on the wall, which object receives that other force? My example shows ___.	Can I draw both contact forces of that pair on the hand? I would change ___ because ___.

Mass and acceleration

ES: masa y aceleración / PT: massa e aceleração

Meaning in my words	How can you tell?
How does mass affect acceleration for the same net force? In my words, this means ___.	Two skaters push off with equal net force magnitudes. Which skater has greater acceleration? I can tell because ___.
Example or sketch	**Non-example and why**
A 10 N net force acts separately on 2 kg and 5 kg carts. Compare their accelerations. My example shows ___.	The lighter cart accelerates more. Does that prove it felt the larger interaction force? I would change ___ because ___.

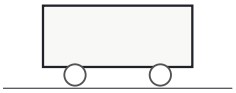
Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

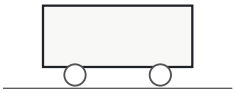
Label both receivers first. Draw equal force arrows on different objects, pointing toward one another for gravity.

Labels you can use: on A · on B · same strength.

Receiver A



Receiver B



Tell a partner

___ pushes on ___. Its partner is ___ pushing on ___.

Two pulls. Two receivers.

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SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Place an interaction pair on its two receivers.

CHOOSE ONE JOB TOGETHER

SIMULATION ENGINEER ☐

Put each gravitational arrow on the body that receives it. Send the animator two corrected object panels.

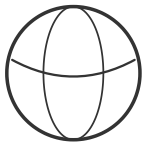
SCIENCE GRAPHICS PRODUCER ☐

Write a split-screen caption naming who pulls whom. Explain why equal force does not require equal acceleration.

Word help: interaction / receiver / equal and opposite

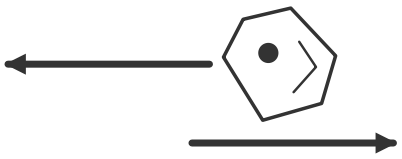
Earth pulls on XJ. A public animation shows both gravitational arrows on XJ, with the caption “they cancel.”

EARTH PANEL



Earth

XJ PANEL



PROPOSAL: “They cancel.”

01 OUR FIRST CALL

Do the two arrows cancel on XJ? Explain using your corrected panels.

___ pulls on ___. That arrow belongs on ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Which arrow belongs in the updated shot? Name its source and receiver.

Now we would say ___ because ___.

Switch speakers. Which object receives each force?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks